

14.6 Testing lack of fit with replicates

Every test and interval so far has assumed that the mean function is right, that $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$. When it is wrong, the residual mean square overestimates the error variance: Example (Residual sum of squares under a wrong mean) showed that $E(\text{SSE}) = \sigma^2(n-r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ for a mean vector $\boldsymbol{\theta}$ outside the model space. Separating noise from systematic error needs an estimate of σ^2 free of the form of the mean, and repeated rows provide one: observations at identical settings differ only by noise. The idea, and the test built on it, go back to Fisher (1922).

14.6.1 Replicates and the cell-means model

Suppose the model matrix has c distinct rows $\mathbf{x}_1^\top, \dots, \mathbf{x}_c^\top$, the i th occurring $n_i \geq 1$ times, with $n = \sum_i n_i$. Index the observations by the distinct row and the replicate, y_{ij} with $j = 1, \dots, n_i$, so that the model is

$$E(y_{ij}) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, c, \quad j = 1, \dots, n_i.$$

Let $\mathbf{Z}$ be the $n \times c$ matrix of group indicators, the model matrix of the one-way layout in which each distinct row has its own mean μ_i . Call two matrices with n rows *alike in row structure* as $\mathbf{X}$ if their rows are equal whenever the corresponding rows of $\mathbf{X}$ are equal.

Lemma 14.6.1. The largest model with the same replicates

- (a) $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$. More generally, every matrix $\mathbf{W}$ with n rows whose rows are constant within the groups satisfies $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$.
- (b) $\mathbf{M}_Z \mathbf{y}$ replaces each observation by its group mean $\bar{y}_i$, and $\mathbf{M} \mathbf{y}$ is constant within groups.

Proof.

- (a) Let $\bar{\mathbf{W}}$ be the $c \times s$ matrix whose i th row is the common row of $\mathbf{W}$ in group i . Then $\mathbf{W} = \mathbf{Z} \bar{\mathbf{W}}$, so $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$. The matrix $\mathbf{X}$ is a special case.
- (b) The first claim is Example (The one-way layout). For the second, $\mathbf{M} \mathbf{y} \in \mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, and the vectors of $\mathcal{C}(\mathbf{Z})$ are constant within groups. ■

So the cell-means model is the largest linear model that treats identical rows identically. Any mean function of the regressors at all, linear or not, produces a mean vector in $\mathcal{C}(\mathbf{Z})$.

Definition 14.6.1. Pure error and lack of fit

With $\mathbf{M}_Z$ the projection onto $\mathcal{C}(\mathbf{Z})$, the **pure error** and **lack-of-fit** sums of squares are

$$\begin{aligned} \text{SSPE} &= \|(\mathbf{I} - \mathbf{M}_Z) \mathbf{y}\|^2 = \sum_{i=1}^c \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2, \\ \text{SSLF} &= \|(\mathbf{M}_Z - \mathbf{M}) \mathbf{y}\|^2 = \sum_{i=1}^c n_i (\bar{y}_i - \hat{y}_i)^2, \end{aligned} \tag{14.6.1}$$

with $n - c$ and $c - r$ degrees of freedom, where $\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ is the common fitted value of group i and $r = \text{rank}(\mathbf{X})$.

The two expressions for each sum of squares agree by [Lemma 14.6.1\(b\)](#): $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$ has entries $y_{ij} - \bar{y}_i$, and $(\mathbf{M}_Z - \mathbf{M})\mathbf{y}$ has entries $\bar{y}_i - \hat{y}_i$, repeated n_i times.

14.6.2 The test

Theorem 14.6.2. Pure-error lack-of-fit test

Let $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$, that is, $E(y_{ij}) = \mu_i$ depends only on the row of $\mathbf{X}$; assume $r < c < n$. Then:

- (a) $\text{SSE} = \text{SSLF} + \text{SSPE}$, and SSLF and SSPE are independent;
- (b) $\text{SSPE}/\sigma^2 \sim \chi^2(n - c)$ for every $\mu_1, \dots, \mu_c$;
- (c) $\text{SSLF}/\sigma^2 \sim \chi^2(c - r, \gamma)$ with

$$\gamma = \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^c n_i (\mu_i - \tilde{\mu}_i)^2, \quad (14.6.2)$$

where $\tilde{\mu}_i$ is the i th group value of $\mathbf{M}\boldsymbol{\theta}$, the least squares fit to the true means;

- (d) the ratio

$$F = \frac{\text{SSLF}/(c - r)}{\text{SSPE}/(n - c)} \sim F(c - r, n - c, \gamma),$$

and $\gamma = 0$ iff $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, that is, iff the model is correct. The test that rejects the model when $F > F_\alpha(c - r, n - c)$ has size α , and its power increases strictly with γ .

Proof. By [Lemma 14.6.1](#), $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$, so by [Theorem 6.5.1](#) $\mathbf{M}_Z - \mathbf{M}$ is the projection onto the orthogonal complement of $\mathcal{C}(\mathbf{X})$ in $\mathcal{C}(\mathbf{Z})$, of rank $c - r$, and $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$ splits into two orthogonal projections. This gives the identity in (a). By [Theorem 4.4.4](#), SSLF/σ^2 and SSPE/σ^2 are independent noncentral chi-squared variables with $c - r$ and $n - c$ degrees of freedom and noncentralities $\|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$. The second is zero because $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$, which proves (b). In the first, $\mathbf{M}_Z\boldsymbol{\theta} = \boldsymbol{\theta}$, so $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\boldsymbol{\theta}$, whose entries are $\mu_i - \tilde{\mu}_i$, repeated n_i times; this is (c). Part (d) follows from [Definition 4.1.2](#), the fact that $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{0}$ iff $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, and [Theorem 4.1.4](#). ■

The test is the F test of the reduced model $\mathcal{C}(\mathbf{X})$ inside the full model $\mathcal{C}(\mathbf{Z})$ ([Theorem 11.1.1](#), [Theorem 4.4.6](#)). What makes it special is the full model, which is chosen not by the analyst but by the replication in the data, and which contains every mean function of the regressors. The residual sum of squares of the working model is split into two rows of an analysis of variance:

Source	Degrees of freedom	Sum of squares	Mean square
Lack of fit	$c - r$	$\sum_i n_i (\bar{y}_i - \hat{y}_i)^2$	$\text{SSLF}/(c - r)$
Pure error	$n - c$	$\sum_i \sum_j (y_{ij} - \bar{y}_i)^2$	$\text{SSPE}/(n - c)$
Residual	$n - r$	SSE	s^2

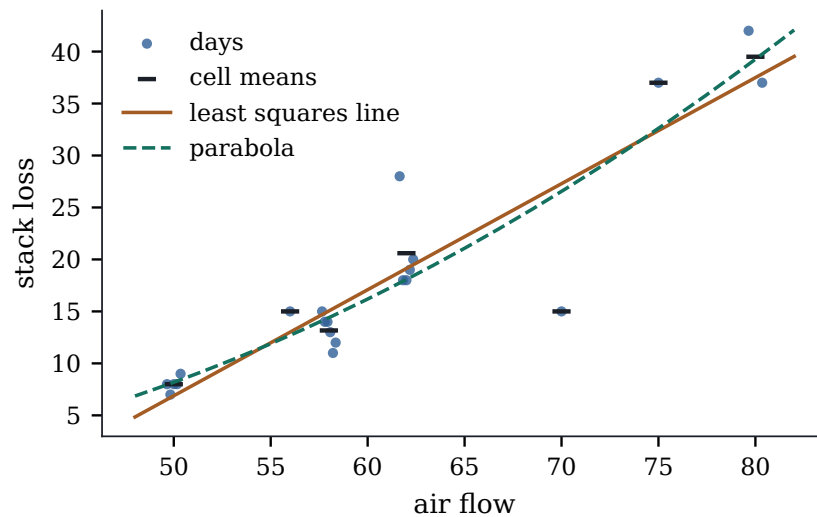


Figure 14.6.1: Stack loss against air flow on 21 days (replicates spread sideways), the seven group means (bars), the least squares line and parabola.

The noncentrality Eq. 14.6.2 is the weighted squared distance between the true group means and their best approximation by the model, in units of σ^2 . It grows with the replication, so replicates buy power twice: through γ and through the pure-error degrees of freedom.

Example. Stack loss and air flow

Brownlee's stack loss data record $n = 21$ days of operation of a plant that oxidizes ammonia. Air flow, the rate of operation, takes only $c = 7$ distinct values, so a straight-line regression of stack loss on air flow can be tested for lack of fit with 14 degrees of freedom for pure error. The line has intercept -44.13 and slope 1.020 , and $R^2 = 0.846$. Its residual sum of squares 319.12 splits into $SSPE = 96.53$ and $SSLF = 222.58$ on 5 degrees of freedom. The mean squares are 44.52 and 6.90, so $F = 6.46$ with p -value 0.0026; the 5% point of $F(5, 14)$ is 2.96. The residual mean square of the line, 16.80, is more than twice the pure-error estimate of σ^2 .

Figure 14.6.1 shows why. The group means do not lie on any smooth curve: the single day at air flow 70 has stack loss 15.0, below the mean 20.6 of the five days at air flow 62. A parabola does not help, with $F = 7.22$ on 4 and 14 degrees of freedom (p -value 0.0023). The mean of stack loss is not a function of air flow alone, which points to the other operating variables, taken up in Example (Is the three-variable model adequate?).

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
x = data["AIRFLOW"].to_numpy()
n = len(y)
levels, groups = np.unique(x, return_inverse=True) # c distinct rows of X
c = len(levels)
Z = np.eye(c)[groups] # cell-means model matrix

def sse(A):
    coef, *_ = np.linalg.lstsq(A, y, rcond=None)
```

```

    return np.sum((y - A @ coef) ** 2)

ss_pe = sse(Z)                                     # pure error, n - c df
for name, X in [("line", np.column_stack([np.ones(n), x])),
                ("parabola", np.column_stack([np.ones(n), x, x ** 2]))]:
    r = np.linalg.matrix_rank(X)
    ss_lf = sse(X) - ss_pe                           # lack of fit, c - r df
    F = (ss_lf / (c - r)) / (ss_pe / (n - c))
    print(f"{name:8s}: SSE {sse(X):7.2f} = SSLF {ss_lf:7.2f} ({c - r} df)"
          f" + SSPE {ss_pe:6.2f} ({n - c} df); F = {F:.2f}, p = {stats.f.sf(F, c -
          r, n - c):.4f}")

```

[Run this code online](#)

14.6.3 When pure error is not pure

The theorem assumes $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$: the mean depends on the observation only through its row of $\mathbf{X}$. If a variable that is not in the model affects the response and varies within the groups of replicates, as water temperature does among the days with equal air flow in the stack loss data, then observations in a group do not share a mean. By Theorem 9.6.1(b), applied to the cell-means model,

$$E\left(\frac{\text{SSPE}}{n - c}\right) = \sigma^2 + \frac{\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2}{n - c},$$

so the “pure” error estimate is inflated, and F becomes a ratio of two noncentral chi-squared variables, a *doubly noncentral F*, which rejects less often than it would with a pure estimate of σ^2 . Two further cautions: the pooled estimate assumes a common variance across groups ([Exercise \(C2\)](#)), and a non-significant result with few degrees of freedom is weak evidence that the model is adequate.

Remark (Designing for a lack-of-fit test). The test needs $c > r$ distinct rows and $n > c$ observations. The design that estimates a straight line most precisely puts half the observations at each end of the range; it has $c = r = 2$ and cannot detect any curvature. A few intermediate settings buy a check of the model at a small cost in precision. Replicates also cap what any model of the regressors can achieve: $R^2 \leq 1 - \text{SSPE}/\text{SST}$ ([Exercise \(B2\)](#)).

14.6.4 Exercises

14.6.4.1 A. Check your understanding

Exercise. A1

A straight line is fitted to $n = 24$ observations at $c = 6$ distinct values of x , each replicated four times. The residual sum of squares is 50.0 and the pure-error sum of squares is 36.0. Complete the analysis of variance table for lack of fit and compute F . Is there evidence of lack of fit at level 0.05, given that $F_{0.05}(4, 18) = 2.93$?

14.6.4.2 B. Practice

Exercise. B1

Show that the least squares estimate $\hat{\boldsymbol{\beta}}$ from all n observations is also the weighted least squares estimate from the c group means, with weights n_i , and that SSLF is the weighted residual sum of squares of that fit.

Solution. By Theorem 6.5.1, $\mathbf{M}\mathbf{y} = \mathbf{M}\mathbf{M}_Z\mathbf{y}$: projecting $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$ is the same as projecting the vector of group means $\mathbf{M}_Z\mathbf{y}$. For a vector constant within groups, $\|\mathbf{M}_Z\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_i n_i(\bar{y}_i - \mathbf{x}_i^\top \mathbf{b})^2$, so minimizing it over $\mathbf{b}$ is weighted least squares on the c means with weights n_i . Its minimum is $\|(\mathbf{M}_Z - \mathbf{M})\mathbf{y}\|^2 = \text{SSLF}$.

Exercise. B2

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Show that every model whose model matrix has the row structure of $\mathbf{X}$ has $R^2 \leq 1 - \text{SSPE}/\text{SST}$, with equality for the cell-means model.

Solution. By Lemma 14.6.1(a) the column space $\mathcal{V}$ of such a model lies in $\mathcal{C}(\mathbf{Z})$, so its residual sum of squares $\|(\mathbf{I} - \mathbf{P}_{\mathcal{V}})\mathbf{y}\|^2$ is at least $\|(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}\|^2 = \text{SSPE}$ (projection onto a smaller space leaves a longer residual). Hence $R^2 = 1 - \text{SSE}/\text{SST} \leq 1 - \text{SSPE}/\text{SST}$, with equality when $\mathcal{V} = \mathcal{C}(\mathbf{Z})$.

Exercise. B3

Show that rows i and j of $\mathbf{M}$ are equal iff rows i and j of $\mathbf{X}$ are equal. So replicates can be read off the hat matrix.

14.6.4.3 C. Going deeper**Exercise. C1**

A straight line is to be tested for lack of fit at the four values $x = 0, 1, 2, 3$, each replicated m times, when the true mean is $\beta_0 + \beta_1 x + \beta_2 x^2$ with $\beta_2/\sigma = 0.5$. Show that $\gamma = m\beta_2^2 \|(\mathbf{I} - \mathbf{M}_4)\mathbf{q}\|^2/\sigma^2$, where $\mathbf{q} = (0, 1, 4, 9)^\top$ and $\mathbf{M}_4$ is the projection for a line on the four points, and find the smallest m for which the test at level 0.05 has power at least 0.8.

Exercise. C2

Suppose the groups have different variances σ_i^2 . Compute $E(\text{SSPE})$ and $E(\text{SSLF})$ under a correct mean model, and show by an example with two large groups of very different sizes and variances that the F test can then reject far more or far less often than α .